

Genomic Variant Analysis Pipeline
HEProAI
BIOINFORMATICS FELLOWSHIP PROGRAM

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INTRODUCTION TO COMPUTATIONAL GENOMICS

Bioinformatics is an interdisciplinary branch of science that integrates biology, computer science, statistics, and information technology to analyze and interpret biological data. The rapid advancement of sequencing technologies has resulted in the generation of massive genomic datasets, making computational analysis essential in modern biological research. Bioinformatics provides powerful computational methods and algorithms that enable researchers to process, store, visualize, and interpret genomic information efficiently.

The field of genomics focuses on studying the complete genetic material of organisms, including the structure, function, and interaction of genes within the genome. Advances in next-generation sequencing (NGS) technologies have significantly improved the speed and accuracy of DNA sequencing, allowing researchers to investigate genomic variations at a large scale. These developments have transformed biological research, clinical diagnostics, and personalized medicine.

Importance of Genomic Variant Analysis

Genomic variant analysis is a critical area of genomics that involves identifying differences between an individual's DNA sequence and a standard reference genome. Genetic variations may occur in different forms, including Single Nucleotide Polymorphisms (SNPs), insertions, deletions (INDELs), duplications, and structural rearrangements. Some of these variants may be harmless, while others may contribute to disease development, genetic disorders, or altered drug responses.

The identification and analysis of genomic variants help researchers understand mutation patterns, disease mechanisms, and hereditary traits. Variant analysis is widely used in medical genetics, cancer research, evolutionary biology, and population genomics. By studying these variations, scientists can identify biomarkers associated with diseases and develop targeted therapeutic strategies.

Role of Sequencing Data in Modern Genomics

Modern sequencing technologies produce large volumes of high-throughput sequencing data that can be analyzed to study genome organization and identify mutations. Sequencing data provides

detailed information about nucleotide composition, genomic rearrangements, and genetic diversity. The analysis of sequencing reads requires computational pipelines capable of handling large datasets with high accuracy and efficiency.

In genomic research, sequencing reads are typically aligned against a reference genome to identify regions of similarity and variation. Alignment tools such as BWA are commonly used to map sequencing reads to the human reference genome. The aligned data is then processed using tools such as SAMtools and BCFtools to identify genomic variants. These computational workflows form the foundation of modern genomic analysis pipelines.

Applications in Disease Detection and Personalized Medicine

Genomic variant analysis has revolutionized healthcare by improving the understanding of genetic diseases and enabling precision medicine approaches. Variants associated with inherited disorders, cancer progression, and drug metabolism can be identified through genomic analysis. This information allows clinicians to make more accurate diagnoses and develop personalized treatment plans based on an individual's genetic profile.

In cancer genomics, variant analysis is used to identify tumor-specific mutations that may influence disease progression and therapeutic response. In pharmacogenomics, genomic data helps predict how patients respond to specific medications, reducing adverse drug reactions and improving treatment effectiveness. Additionally, genomic variant analysis plays an important role in prenatal screening, infectious disease research, and population genetics studies.

Scope of the Present Study

The present study focuses on implementing a simplified genomic variant analysis pipeline using bioinformatics tools within the Google Colab environment. Simulated FASTQ sequencing reads generated from chromosome 22 of the GRCh38 human reference genome were analyzed to identify genomic variants. The workflow included sequence alignment, BAM file processing, variant calling, filtering, visualization, and report generation.

The project demonstrates the practical application of widely used bioinformatics tools such as BWA, SAMtools, BCFtools, Pandas, and Matplotlib in genomic data analysis. The study also

provides hands-on experience in handling sequencing datasets and understanding the computational methods involved in genomic variant detection and interpretation.

AIM OF THE PROJECT

The primary aim of this project is to design and implement a genomic variant analysis workflow using widely used bioinformatics tools within the Google Colab environment.

Project Goals

- To preprocess and manage sequencing datasets
- To align sequencing reads with the GRCh38 human reference genome
- To perform genomic variant identification
- To classify and visualize detected mutations
- To prepare an organized genomic analysis report

SOFTWARE AND TECHNOLOGIES IMPLEMENTED

Software/Technology	Function
Python	Scripting and data analysis
BWA	Read alignment
SAMtools	SAM/BAM file processing
BCFtools	Variant identification
Pandas	Data manipulation
Matplotlib	Graph generation
Google Colab	Cloud-based computational platform

1. DATASET SPECIFICATIONS

File Format: FASTQ

The sequencing dataset used in this project was stored in FASTQ format, which is widely used in next-generation sequencing analysis. FASTQ files contain both nucleotide sequence information and corresponding quality scores for each sequencing read. These quality scores indicate the reliability and accuracy of each detected nucleotide base. The format is highly suitable for genomic analysis because it combines sequence and quality information within a single file. FASTQ files serve as the primary input for downstream processes such as alignment and variant detection.

Reference Genome: GRCh38 Chromosome 22

The reference genome used in this study was chromosome 22 from the GRCh38 human genome assembly. GRCh38 is one of the most widely accepted reference genomes used in genomic research and clinical analysis. Chromosome 22 was selected because of its relatively smaller size, making it computationally efficient for experimental studies and learning purposes. The reference genome acts as a standard DNA sequence against which sequencing reads are aligned. Comparing sequencing reads with the reference genome enables the identification of genomic variations such as SNPs and INDELs.

Data Type: Simulated Sequencing Reads

The dataset consisted of simulated sequencing reads generated computationally from chromosome 22 of the human reference genome. Simulated reads mimic real sequencing data produced by next-generation sequencing platforms. The use of simulated data provides a controlled environment for testing genomic analysis workflows without involving real patient data. It also simplifies the validation of detected variants and reduces computational complexity during analysis. These sequencing reads were later aligned to the reference genome for variant identification.

Mutation Strategy: Artificial Variants Introduced for Analysis Validation

Artificial mutations were intentionally introduced into the sequencing dataset to ensure successful detection of genomic variants during analysis. These mutations included nucleotide substitutions, insertions, and deletions that simulated real genomic variations. Introducing artificial variants helped validate the accuracy and efficiency of the bioinformatics workflow used in this project. The strategy allowed the variant calling tools to detect known mutations and evaluate pipeline performance effectively. This approach provided practical understanding of genomic variant identification and analysis techniques.

Dataset Summary

The dataset used in this project consisted of simulated FASTQ sequencing reads generated from chromosome 22 of the GRCh38 human reference genome. Artificial genomic variants were introduced to validate the efficiency of the variant detection workflow. The sequencing data was processed through multiple bioinformatics stages including alignment, BAM processing, variant calling, filtering, and visualization. The study demonstrated the practical application of bioinformatics tools in genomic data analysis. The dataset provided valuable hands-on experience in understanding sequencing workflows and genomic variant interpretation.

2. METHODOLOGICAL FRAMEWORK

Workflow Pipeline

FASTQ File

↓

Quality Cleaning

↓

Sequence Alignment

↓

BAM Conversion & Sorting

↓

Variant Calling

↓

Variant Filtering

↓

Variant Analysis

↓

Report Generation

Workflow Explanation

Data Upload

FASTQ sequencing files were uploaded into the Google Colab environment for genomic data processing.

Read Trimming

Low-quality reads and unwanted sequencing regions were removed to improve alignment quality and analytical accuracy.

Sequence Alignment

Processed sequencing reads were aligned against the GRCh38 chromosome 22 reference genome using the BWA alignment algorithm.

BAM Conversion and Sorting

SAM alignment files were converted into BAM format, followed by sorting and indexing using SAMtools.

Variant Calling

Genomic variants were identified using BCFtools through mpileup generation and variant calling procedures.

Variant Filtering

Low-confidence variants were filtered based on quality score thresholds to retain reliable variant calls.

Variant Analysis and Reporting

Filtered variants were analyzed using Python libraries such as Pandas and visualized using Matplotlib for report generation.

3. ACTIVITY 1: VARIANT DETECTION

Environment Setup

Installation Commands

```
!apt-get install -y bwa samtools bcftools
```

Computational Environment

Google Colab was used as the cloud-based execution platform for running the genomic analysis workflow.

Reference Genome Preparation

Procedures Performed

- Downloading GRCh38 chromosome 22 reference genome
- Indexing the reference genome using BWA

Example Command

```
!bwa index chr22.fa
```

Read Alignment

Alignment Command

```
!bwa mem chr22.fa sample_trimmed.fastq > aligned.sam
```

Description

The BWA-MEM alignment algorithm was used to map sequencing reads against the reference genome, producing an aligned SAM file for downstream analysis.

BAM File Processing

Processing Steps

- Conversion of SAM to BAM format
- Sorting of BAM files
- Indexing of BAM files

Commands Used

```
!samtools view -Sb aligned.sam > aligned.bam
```

```
!samtools sort aligned.bam -o aligned_sorted.bam
```

```
!samtools index aligned_sorted.bam
```

Variant Calling Procedure

Variant Detection Method

The mpileup function and BCFtools variant caller were used to identify SNPs and INDELs from aligned sequencing reads.

Command Used

```
!bcftools call -mv raw_variants.vcf -Ov -o variants.vcf
```

Variant Quality Filtering

Filtering Strategy

Variants with low-quality scores were excluded to improve the accuracy and reliability of downstream analysis.

4. ACTIVITY 2: VARIANT ANALYSIS AND REPORTING

Loading VCF Data

The VCF file was converted into a Pandas DataFrame to simplify genomic data analysis and manipulation.

Python Code

```
import pandas as pd

vcf = pd.read_csv("filtered_variants.vcf", comment='#', sep='\t',
                  header=None)

vcf.columns = ["CHROM", "POS", "ID", "REF", "ALT",
               "QUAL", "FILTER", "INFO"]
```

Variant Classification

Types of Variants

- SNPs (Single Nucleotide Polymorphisms)
- INDELs (Insertions and Deletions)

Classification Code

```
df["Variant_Type"] = df.apply(
    lambda x: "SNP" if len(x["REF"]) == 1 and len(x["ALT"]) == 1 else "INDEL",
    axis=1
)
```

Data Visualization

Variant Distribution Graph

```
import matplotlib.pyplot as plt

df["Variant_Type"].value_counts().plot(kind="bar")

plt.title("Variant Type Distribution")
plt.xlabel("Variant Type")
plt.ylabel("Count")
plt.show()
```

5. REPORT GENERATION

Reporting Process

- Exporting analyzed data into CSV format
- Generating summarized genomic reports

Export Command

```
vcf.to_csv("variant_report.csv", index=False)
```

6. RESULTS AND OUTPUTS

RESULTS AND ANALYSIS SUMMARY

The genomic variant analysis pipeline was successfully executed using simulated FASTQ sequencing data aligned against chromosome 22 of the GRCh38 reference genome. The workflow included sequence alignment, BAM processing, variant calling, filtering, variant classification, and graphical visualization using bioinformatics tools within the Google Colab environment. The analysis generated a total of 14 high-confidence genomic variants after quality filtering. All identified variants passed the filtering criteria and were retained for downstream analysis

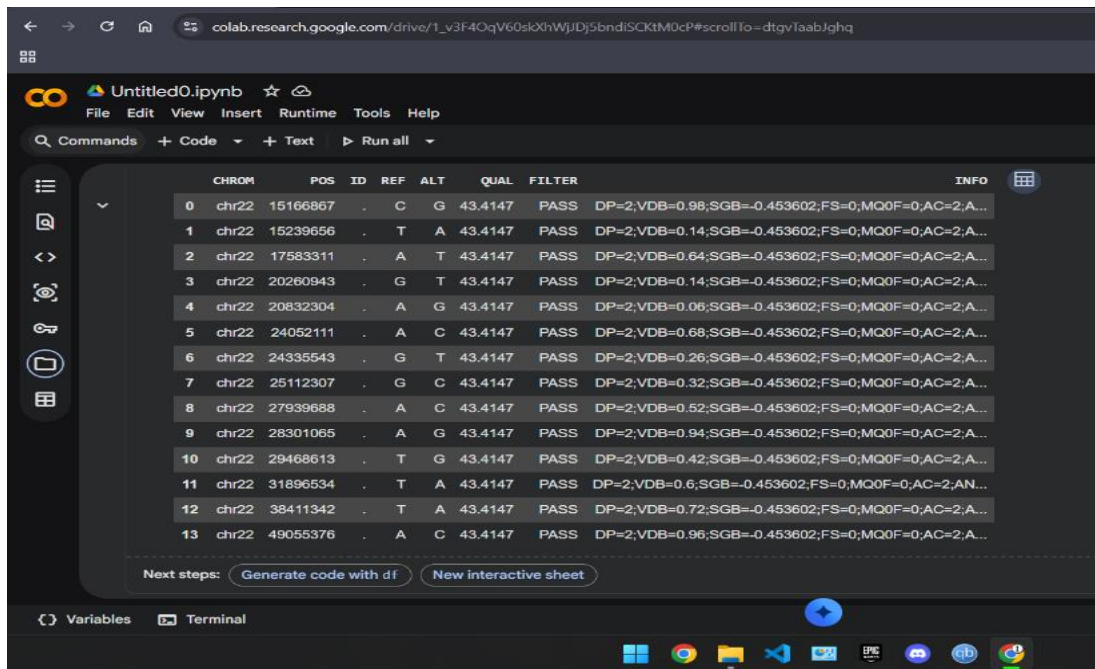
Variant report

CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	Variant_1
chr22	15166867	.	C	G	43.4147	PASS	DP=2;VDB=0.98;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=55	SNP
chr22	15239656	.	T	A	43.4147	PASS	DP=2;VDB=0.14;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	17583311	.	A	T	43.4147	PASS	DP=2;VDB=0.64;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	20260943	.	G	T	43.4147	PASS	DP=2;VDB=0.14;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=58	SNP
chr22	20832304	.	A	G	43.4147	PASS	DP=2;VDB=0.06;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	24052111	.	A	C	43.4147	PASS	DP=2;VDB=0.68;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=50	SNP
chr22	24335543	.	G	T	43.4147	PASS	DP=2;VDB=0.26;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	25112307	.	G	C	43.4147	PASS	DP=2;VDB=0.32;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	27939688	.	A	C	43.4147	PASS	DP=2;VDB=0.52;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	28301065	.	A	G	43.4147	PASS	DP=2;VDB=0.94;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	29468613	.	T	G	43.4147	PASS	DP=2;VDB=0.42;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	31896534	.	T	A	43.4147	PASS	DP=2;VDB=0.6;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	38411342	.	T	A	43.4147	PASS	DP=2;VDB=0.72;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP
chr22	49055376	.	A	C	43.4147	PASS	DP=2;VDB=0.96;SGB=-0.453602;FS=0;MQOF=0;AC=2;AN=2;DP4=0,0,2,0;MQ=60	SNP

Variant Summary Table

Metric	Value
Total Variants	14
SNP Count	14
INDEL Count	0

Alignment table



The screenshot shows a Google Colab notebook interface. The main content area displays a table with 14 rows of variant calling data. The table has columns for CHROM, POS, ID, REF, ALT, QUAL, FILTER, and INFO. The data is as follows:

	CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO
0	chr22	15166867	.	C	G	43.4147	PASS	DP=2;VDB=0.98;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
1	chr22	15239656	.	T	A	43.4147	PASS	DP=2;VDB=0.14;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
2	chr22	17583311	.	A	T	43.4147	PASS	DP=2;VDB=0.64;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
3	chr22	20260943	.	G	T	43.4147	PASS	DP=2;VDB=0.14;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
4	chr22	20832304	.	A	G	43.4147	PASS	DP=2;VDB=0.06;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
5	chr22	24052111	.	A	C	43.4147	PASS	DP=2;VDB=0.68;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
6	chr22	24335543	.	G	T	43.4147	PASS	DP=2;VDB=0.26;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
7	chr22	25112307	.	G	C	43.4147	PASS	DP=2;VDB=0.32;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
8	chr22	27939688	.	A	C	43.4147	PASS	DP=2;VDB=0.52;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
9	chr22	28301065	.	A	G	43.4147	PASS	DP=2;VDB=0.94;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
10	chr22	29468613	.	T	G	43.4147	PASS	DP=2;VDB=0.42;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
11	chr22	31896534	.	T	A	43.4147	PASS	DP=2;VDB=0.6;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
12	chr22	38411342	.	T	A	43.4147	PASS	DP=2;VDB=0.72;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...
13	chr22	49055376	.	A	C	43.4147	PASS	DP=2;VDB=0.96;SGB=-0.453602;FS=0;MQ0F=0;AC=2,A...

Below the table, there are two buttons: "Generate code with df" and "New interactive sheet". The bottom of the notebook shows a terminal and a variables panel.

Variant Detection Results

The variant calling process identified a total of 14 genomic variants within the analyzed sequencing dataset. All detected variants were classified as Single Nucleotide Polymorphisms (SNPs), while no insertions or deletions (INDELs) were observed. This indicates that the detected genomic alterations involved substitutions of single nucleotide bases rather than structural sequence changes.

The identified variants were distributed across multiple genomic positions on chromosome 22. Each detected variant contained information including chromosome number, genomic position,

reference allele, alternate allele, quality score, filter status, and variant annotation details. The sequencing reads aligned successfully against the reference genome, allowing accurate identification of nucleotide substitutions.

Quality Analysis of Variants

The detected variants showed a consistent quality score of approximately 43.41, indicating high-confidence variant calls. Quality scores in genomic analysis represent the probability that the detected variants are genuine and not caused by sequencing errors. The relatively high QUAL values suggest that the variant calling pipeline performed effectively and produced reliable results.

All variants were marked with a “PASS” status under the FILTER column, confirming that they successfully satisfied the applied quality filtering conditions. This demonstrates that the selected filtering threshold efficiently removed low-confidence variants while retaining reliable genomic alterations for analysis.

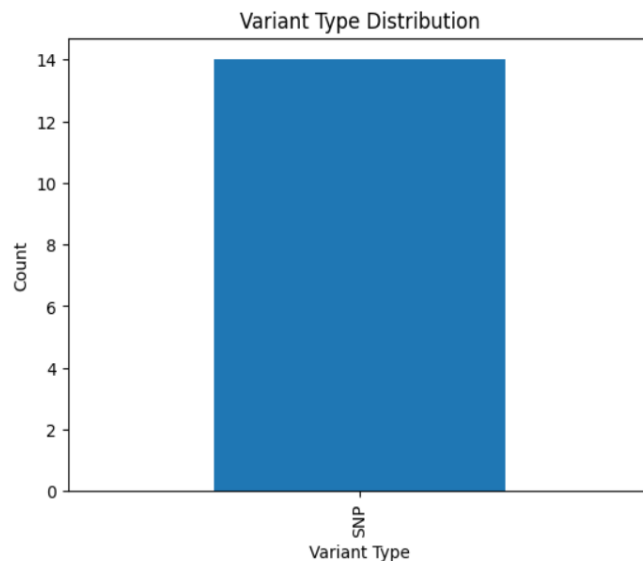
Variant Classification Analysis

The classification analysis confirmed that all 14 detected variants belonged to the SNP category. SNPs represent the most common type of genomic variation in DNA sequences and involve substitution of a single nucleotide base. Examples observed in the dataset included nucleotide substitutions such as:

- Cytosine (C) replaced by Guanine (G)
- Thymine (T) replaced by Adenine (A)
- Adenine (A) replaced by Thymine (T)
- Guanine (G) replaced by Thymine (T)

The absence of INDELs suggests that the introduced mutations within the simulated dataset primarily consisted of single nucleotide substitutions rather than insertion or deletion events.

Graph Summary



The bar chart illustrates the distribution of variant types identified during genomic variant analysis. The results indicate that all detected variants in the dataset were classified as Single Nucleotide Polymorphisms (SNPs), with a total count of 14 variants. No INDELs (insertions or deletions) were observed in the analyzed sequencing data. The predominance of SNPs suggests that single nucleotide substitutions were the primary form of genetic variation detected in this study. This visualization confirms the successful execution of the variant calling and classification workflow using the bioinformatics pipeline.

7. CHALLENGES ENCOUNTERED

Difficulties Faced During Analysis

- Alignment-related errors occurred due to mismatches between file names and reference genome inputs, which required correction before successful sequence alignment.
- BAM indexing complications were encountered when improperly generated or unsorted BAM files prevented successful indexing operations.
- Variant calling syntax issues occurred because of incorrect command formatting and parameter placement, which had to be corrected for proper execution.
- Handling large genomic datasets was challenging because genomic files required significant processing time and computational resources during analysis.
- Memory limitations in the Google Colab environment caused the runtime session to refresh automatically, resulting in loss of uploaded files and outputs, requiring the entire workflow to be repeated from the beginning.
- File naming inconsistencies between generated outputs and command inputs created errors during processing, which were resolved by correcting file names and paths.

8. LEARNING OUTCOMES

Skills and Knowledge Acquired

- Learned genomic data preprocessing techniques
- Understood sequence alignment methodologies
- Performed genomic variant identification
- Gained practical experience with bioinformatics tools
- Worked with GRCh38 human genome datasets
- Improved data visualization and scientific reporting skills

9. CONCLUSION

This project successfully demonstrated a simplified genomic variant analysis workflow using the GRCh38 human reference genome and bioinformatics tools within the Google Colab environment. The pipeline enabled sequencing read alignment, variant detection, filtering,

analysis, and visualization. The project provided practical exposure to genomic data analysis techniques and highlighted the growing importance of bioinformatics in modern genomics and personalized medicine.

10. REFERENCES

- [SAMtools Documentation](#)
- [BCFtools Documentation](#)
- [GRCh38 Reference Genome](#)
- [BioPython Documentation](#)
- [Google Colab Documentation](#)